

3.8 Maxima and Minima of Functions of Two Variables Worksheet

1. True or False: Every function has at least one local maximum.

2. Suppose that $z = f(x, y)$ is a differentiable function and $D = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2$. Determine if each of the following are True or False.
 - If f has a local maximum or minimum at (a, b) , then it always satisfies $f_x(a, b) = f_y(a, b) = 0$.
 - If $f_x(a, b) = f_y(a, b) = 0$, then f has a local maximum or minimum at (a, b) .
 - If f has a local minimum at (a, b) , then $f_{xx}(a, b) > 0$ and $D > 0$.

3. Find all critical points of the function $f(x, y) = yxe^{-(x^2+y^2)}$.

4. Find the critical points of $f(x, y) = 5 + 6x - x^2 + xy - y^2$ and classify them as local maxima, local minima, or saddles points.

5. Let $f(x, y)$ be a function of two variables and suppose (a, b) is a critical point of f . Furthermore, suppose:

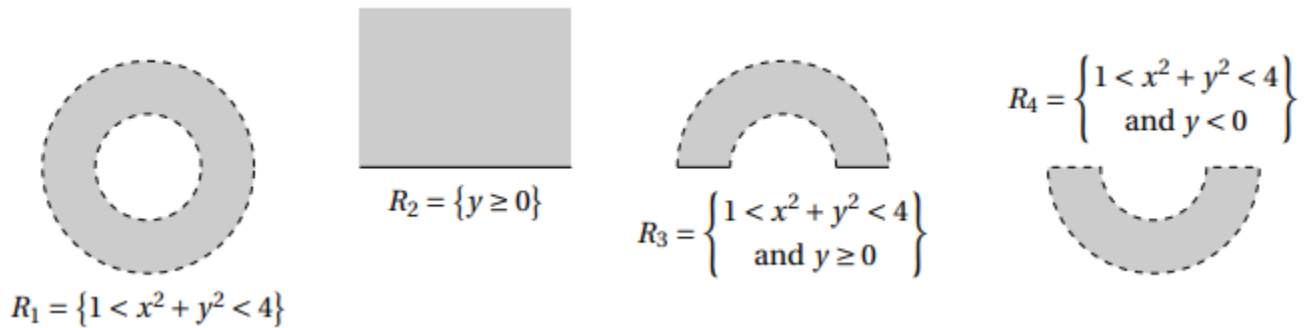
- $f_{xx}(a, b) = 1$
- $f_{xy}(a, b) = 2$
- $f_{yy}(a, b) = 7$

Is $f(a, b)$ a local minimum, local maximum, or neither?

6. Find the local maximum and minimum values and saddle points of the function $f(x, y) = x^3 - 12xy + 8y^3$.

7. Find the critical points of $f(x, y) = x^4 + 4xy + xy^2$ and classify them as local maxima, local minima, or saddles points.

8. Consider the following four regions in the plane.



a) Classify each region as either open, closed, or neither.

b) Classify each region as either bounded or unbounded.

9. Let $f(x, y)$ be a differentiable function on the disk $D = \{(x, y) | x^2 + y^2 = 100\}$ in $\mathbb{R}^2$, where

- $f(x, y) = 20$ for every point (x, y) on the circle $x^2 + y^2 = 100$.
- $f(0, 0) = 5$
- f has only one critical point, which is at $(1, 2)$.

Decide which of the four statements below is true.

- A. $f(1, 2) > 5$
- B. $f(1, 2) < 5$
- C. $f(1, 2) = 5$
- D. The relationship between $f(1, 2)$ & 5 cannot be determined from the given information.

10. Find the absolute maximum and minimum values of $f(x, y) = x^2 + 4y^2 - 2x^2y + 4$ on the rectangle given by $D = \{(x, y) \mid -1 \leq x \leq 1 \text{ \& } -1 \leq y \leq 1\}$.

11. Find the absolute maximum and minimum values of $f(x, y) = 2x^2 - y^2 + 6y$ on region

$$D = \{(x, y) \mid x^2 + y^2 \leq 16\}.$$

12. Consider again the function $f(x, y) = 2x^2 - y^2 + 6y$ on region $D = \{(x, y) \mid x^2 + y^2 \leq 16\}$ as in #11.

a) Why would it be inconvenient to divide the boundary of D into two curves with formulas $y = \pm\sqrt{16 - x^2}$?

b) What would the function $f(x, y) = 2x^2 - y^2 + 6y$ need to look like so that dividing the boundary of D into two curves with formulas $y = \pm\sqrt{16 - x^2}$ *would be* convenient to use?

13. Find the absolute maximum and minimum values of $f(x, y) = x^3 - 4xy^2$ on the triangle with vertices $(0, 0)$, $(4, 2)$, & $(-2, 2)$.